

UR/CST: Mathematics for Engineers I
Year I Computer Engineering, EPE and ETE

Continuous Assessment Test (/40 marks)

Time: 2 hours

October 31, 2024

Q.1. (/3 marks). Find c such that $f(x) = \begin{cases} x & \text{if } x \leq 1 \\ 3x + 2 & \text{if } 1 < x < 4 \\ 5x + c & \text{if } 4 \leq x \end{cases}$ is continuous at $x = 4$.

Q.2. (/4 marks). Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

Q.3. (/3 marks) Find equations of the tangent and normal to $y = \frac{x^2 + 1}{x - 1}$ at $x = 2$.

Q.4. (/4 marks). Evaluate $\lim_{x \rightarrow 0} \frac{3x + |x|}{7x - 5|x|}$ if it exists.

Q.5. (/3 marks). What one-sided limits does $f(x) = \sqrt{1 - x^2}$ have at $x = 1$ and $x = -1$?

Q.6. (/7 marks). Evaluate

1. $A = \int_0^1 \frac{1}{\sqrt{1-x}} dx$

3. $C = \int_0^2 2x(1+x^2)^3 dx$

2. $B = \int \sin^2 x \cdot \cos^3 x dx$

Q.7. (/5 marks). Given that $f(x) = 8x - x^2$,

1. Is the function $f(x)$ even or odd.

2. Find the maximum and the minimum values of $f(x)$ and state where these values occur.

3. On which intervals, the function $f(x)$ is decreasing or increasing.

Q.8. (/3 marks). Find two numbers whose sum is 20 and product is: a) maximum and b) minimum.

Q.9. (/5 marks) Evaluate

a. $\lim_{x \rightarrow 2^-} \frac{-5}{x-2}$ and b. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$

Q.10. (/3 marks) Evaluate

$$\frac{d}{dx} \left[\int_1^{x^2} \frac{\sin t}{t} dt \right]$$

GOOD LUCK